

Understanding Customer Reviews Through Sentiment Analysis

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Imagine you are working with a restaurant company that operates multiple locations and receives thousands of customer reviews online. While each review includes a star rating, the written comments often explain what customers actually liked or disliked in more detail. However, with so many reviews coming in, it is difficult for the company to read and interpret all of them manually.

Because of this, the company is looking for a way to automatically analyze review text to better understand overall customer sentiment. They want to know not just how customers rate their experience, but what patterns appear in the language of those reviews. Being able to quickly identify whether feedback is positive, neutral, or negative would allow them to make more informed decisions about operations, customer experience, and areas for improvement.

Your task is to help solve this problem by building a simple model that can read restaurant reviews and classify the sentiment expressed in each one. Using only the text of the reviews, you will develop a model that predicts whether a review reflects a positive, neutral, or negative experience. This will allow the company to move beyond simple star ratings and better understand customer feedback at a larger scale.

At the end of this case study, you will produce a working sentiment classification model along with a clear explanation of how it performs and what it reveals about customer reviews. Your work will show how data can be used to better understand real-world customer behavior.